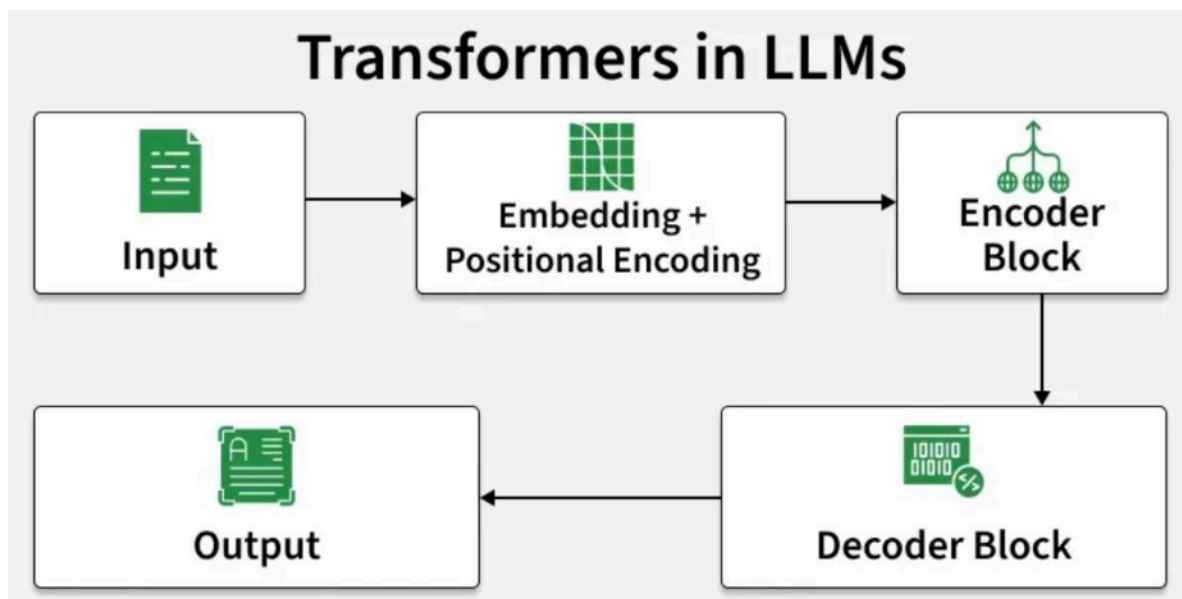




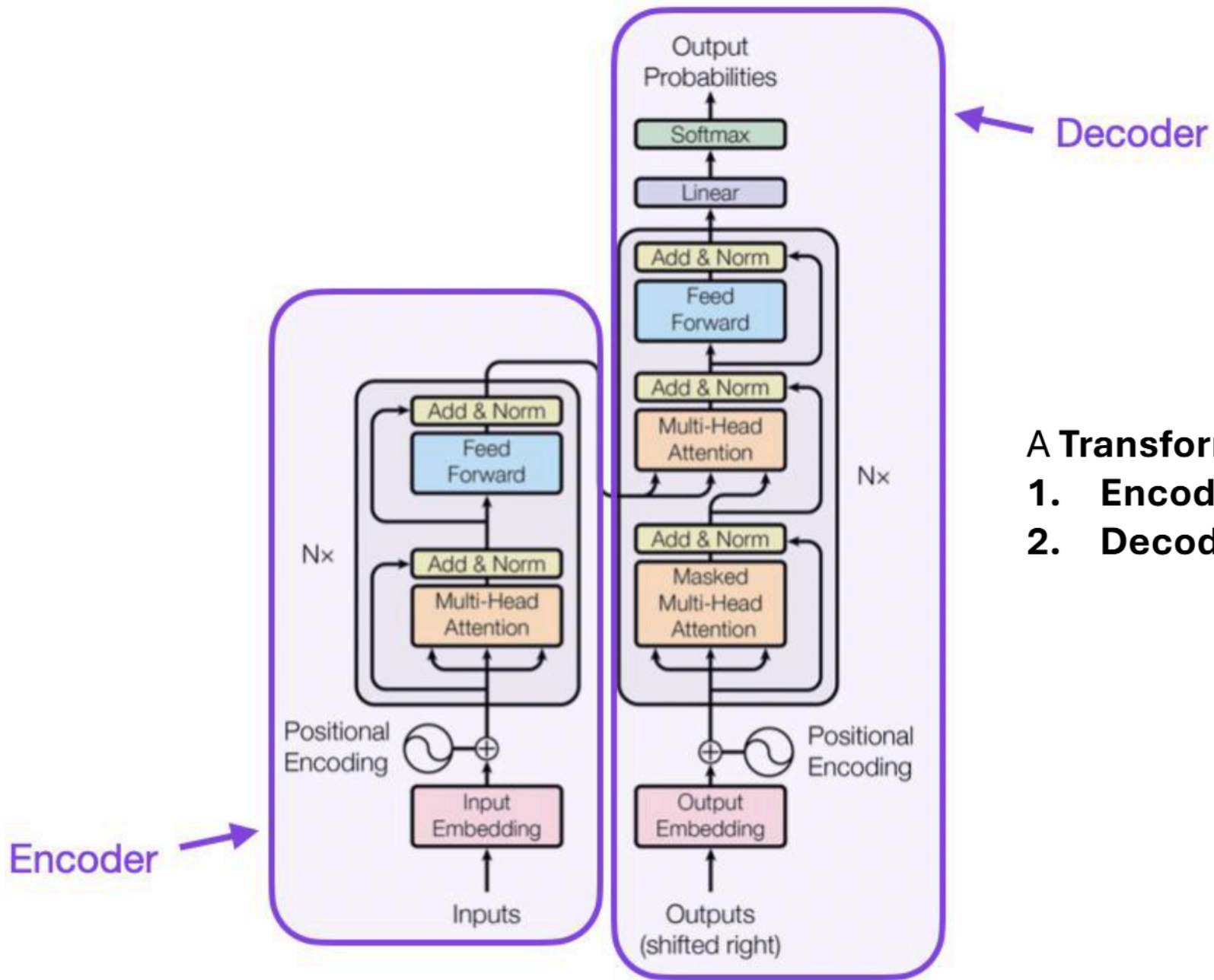
Essential Concepts in Generative AI

*Understanding core concepts
before getting started*

What is a large language model?



- **Large Language Models (LLMs)** are advanced AI systems built on **deep neural networks** designed to process, understand and generate human-like text.
- By using massive datasets and billions of parameters, LLMs have transformed the way humans interact with technology.
- It learns patterns, grammar and context from text and can answer questions, write content, translate languages and many more.



A **Transformer** is made up of **two main parts**:

1. **Encoder** → understands the input
2. **Decoder** → generates the output

Figure 1: The Transformer - model architecture.

Transformers architecture

Each part (Encoder/Decoder) has stacked layers, and every layer has:

1. Self-Attention Layer

- Learns how each word relates to others in the same sentence
- Uses Q (Query), K (Key), and V (Value) vectors to compute attention scores
- Example: In “The cat sat on the mat,” the word “cat” will attend to “sat” and “mat”.

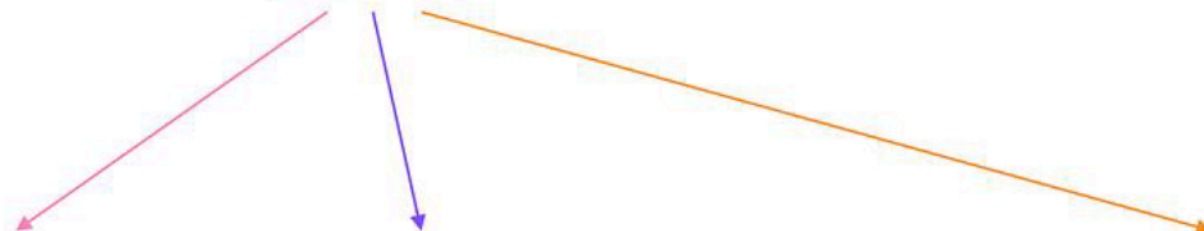
2. Feed-Forward Neural Network

- After attention finds relationships, this small network learns deeper patterns.
- Works independently on each word’s representation.

3. Add & Norm

- Keeps training stable by adding the original input (residual connection) and normalizing values for consistency.

What does **Generative Pre-trained Transformer (GPT)** mean



Generative

Means “next word prediction.”

As just described.

Pre-trained

The LLM is pretrained on massive amounts of text from the internet and other sources.

See next slide.

Transformer

The neural network architecture used (introduced in 2017).

Won't go into more details here.

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What We'll Cover



Core Concepts in Generative AI



Retrieval & Knowledge Augmentation Concepts



Agentic AI Concepts



Evaluation & Safety Concepts

Core Concepts in Generative AI

Hallucination

Hallucination



Definition of Hallucination

Hallucination occurs when AI produces factually incorrect but plausible-sounding information.

Examples of Hallucination

Examples include AI falsely reporting events or citing non-existent sources.

Role in Large Language Models

LLMs generate text based on patterns, which can produce convincing but false content.

Prompt Engineering

Prompt Engineering

Definition of Prompt Engineering

Prompt engineering is the art and science of crafting inputs to optimize AI model outputs effectively.

Importance in Language Models

Large language models are highly sensitive to prompt phrasing, making good prompts essential for better results.

Practical Use Case

Customer service chatbots utilize prompt engineering to deliver consistent and helpful responses to users.



Context Window

Context Window

Context Window Definition

The context window is the maximum text length an AI model can remember and analyze at once.

Role in Language Models

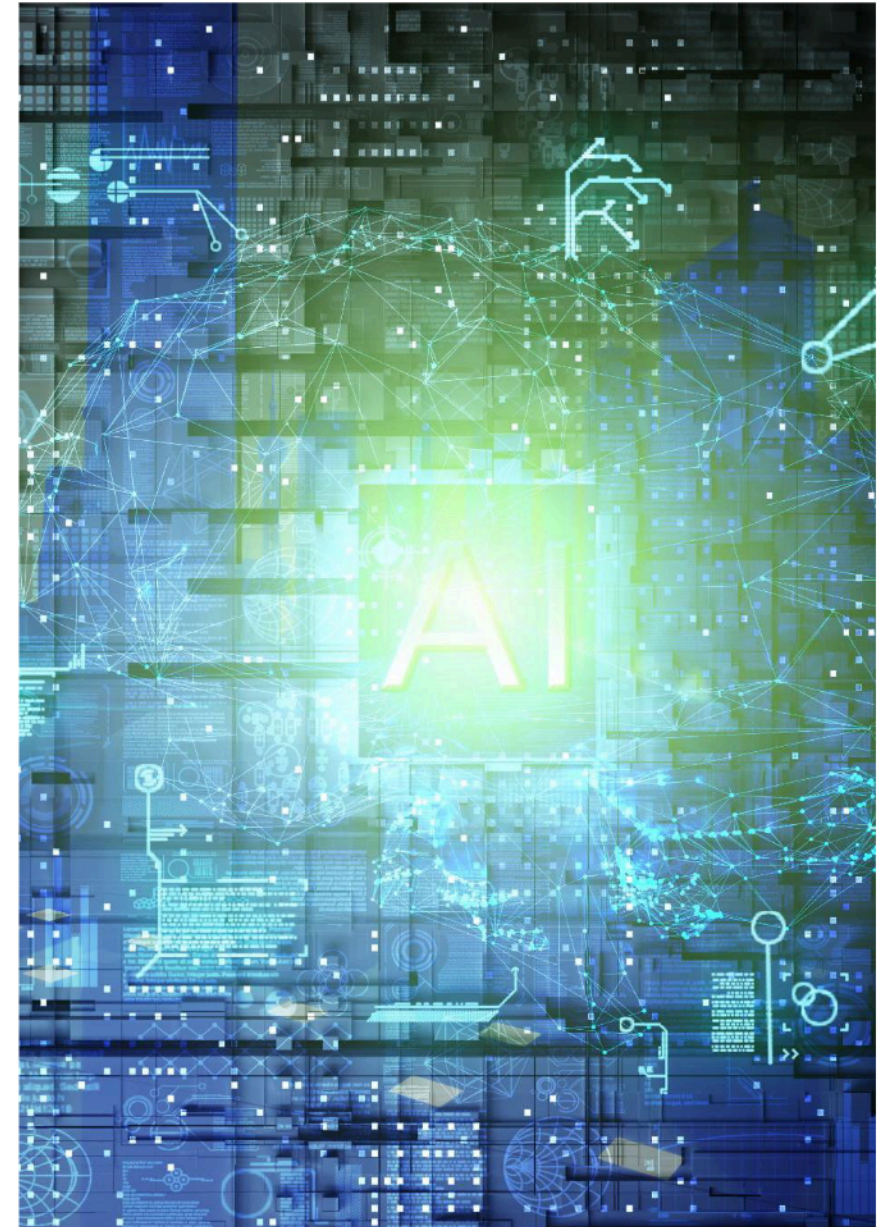
Context window size determines how much conversation or document text an AI can process simultaneously.

Application in Legal Analysis

Large context windows enable AI to analyze entire legal contracts effectively and accurately.

Example

GPT-4 can remember about 32,000 tokens (~24,000 words) of conversation history



Token

Token

Definition

The basic units that AI models use to process text - roughly 3-4 characters or parts of words

Role in Language Models

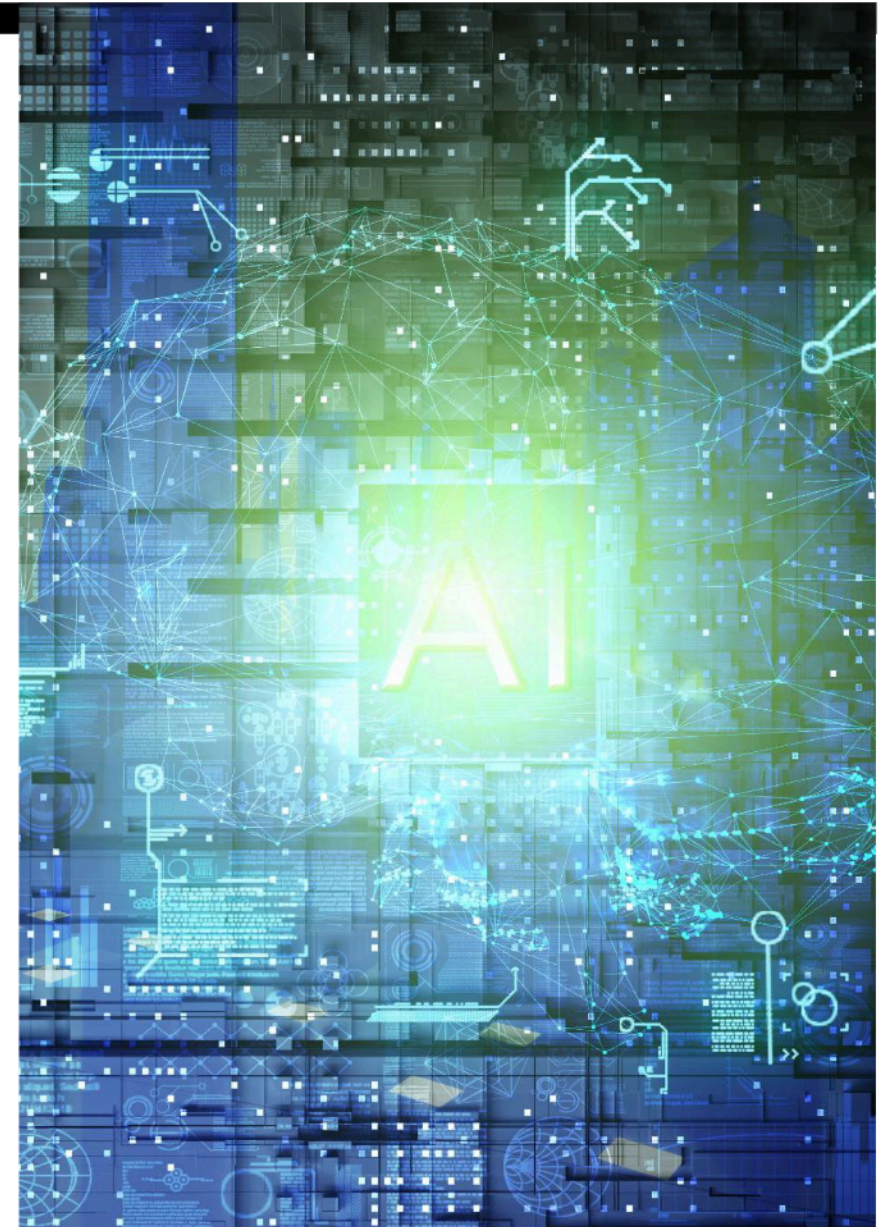
LLMs process and generate text token by token, affecting cost and processing speed

Application in Legal Analysis

API pricing is often based on token usage, so understanding tokens helps control costs

Example

The word 'artificial' might be split into tokens: 'art', 'ific', 'ial'



Temperature

Temperature



Definition

A setting that controls how creative or predictable AI responses are (0 = predictable, 2 = very creative)

Examples of Hallucination

Low temperature: 'The sky is blue.' High temperature: 'The sky resembles a sapphire canvas.'

Role in Large Language Models

Controls randomness in word selection - higher values increase diversity but may reduce accuracy

Use Case in Medical Diagnosis

Creative writing uses high temperature; factual reporting uses low temperature

***Few-Shot /Zero-Shot
/One-Shot Learning***

Few-Shot / Zero-Shot / One-Shot Learning



Definition

Teaching AI by providing examples: Zero (no examples), One-Shot (1 example), Few-Shot (multiple examples)

Examples

Zero-shot: 'Classify sentiment.' Few-shot: 'Happy=Positive, Sad=Negative. Now classify: Excited.'

Role in Large Language Models

Allows models to perform new tasks without retraining by learning from examples in the prompt

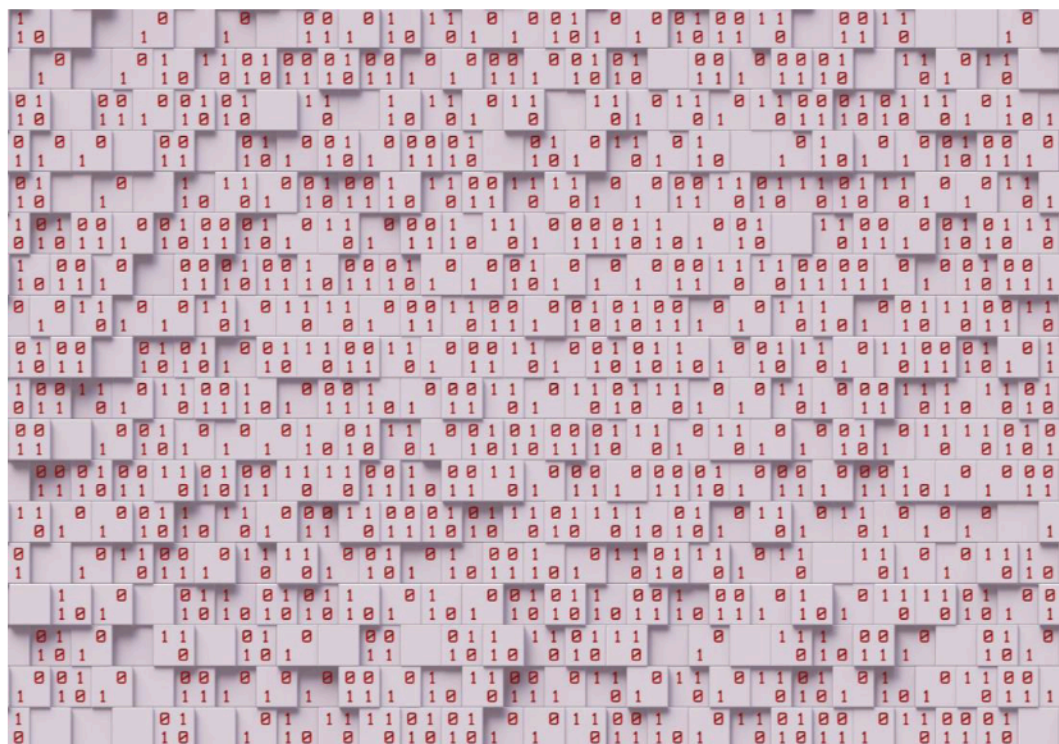
Use Case in Medical Diagnosis

Email classification systems learn to categorize emails with just a few examples

Embeddings and Chain of Thought Reasoning

Embeddings

Embeddings



Definition of Embeddings

Embeddings are numerical representations of words capturing their meanings in vector form.

Semantic Similarity Example

Words like 'king' and 'queen' have similar embeddings, showing related meanings, unlike 'king' and 'banana'.

Role in Large Language Models

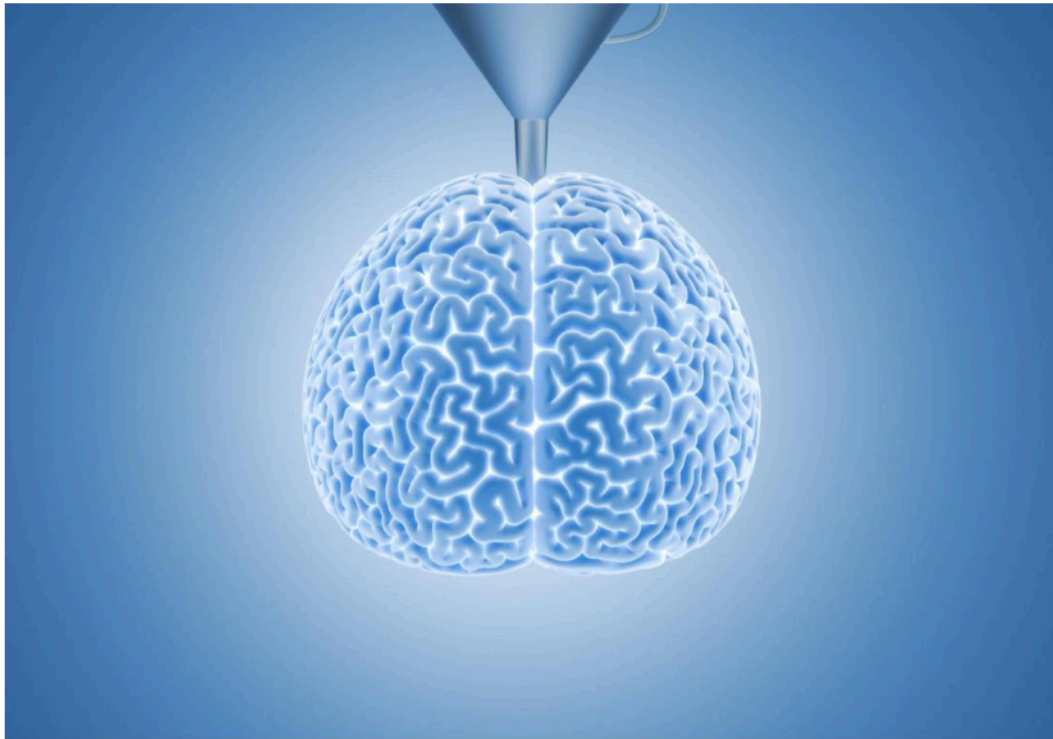
Embeddings enable LLMs to understand concept relationships and perform semantic searches effectively.

Practical Use Cases

Recommendation systems use embeddings to suggest products or content based on similarity and relevance.

Chain of thought

Chain of Thought (CoT): Step-by-Step Reasoning



Definition of CoT

Chain of Thought is a method where AI describes its reasoning steps to improve answer clarity.

Example Explanation

Instead of just answering '15,' the AI shows: '5 apples + 3 apples = 8 apples, then $8 + 7 = 15$ apples total'

Role in Large Language Models

CoT enhances accuracy in complex problems by making AI's reasoning explicit and transparent.

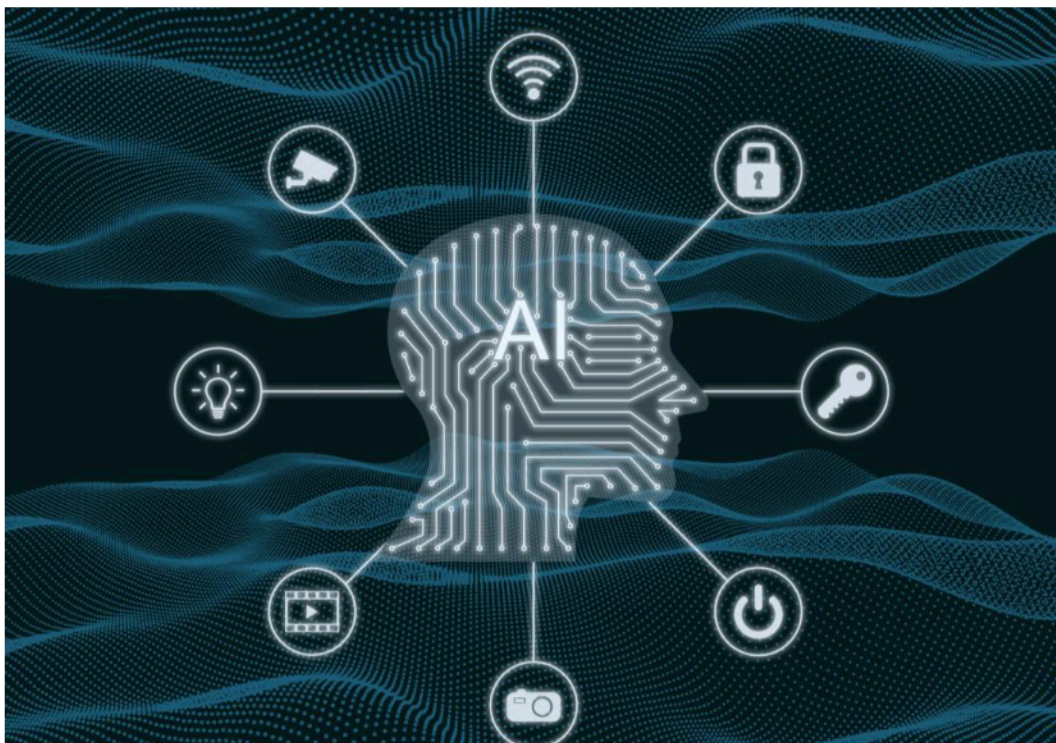
Educational Use Cases

Math tutoring apps use CoT to support students in understanding problem-solving steps effectively.

Retrieval & Knowledge Augmentation Concepts

Grounding

Grounding: Connecting AI to Real Information



Definition of Grounding

Grounding connects AI responses to real, verifiable information sources to reduce hallucinations.

Example Application

Instead of guessing stock prices, AI retrieves current market data before responding.

Role in Large Language Models

Grounding anchors model outputs to factual sources beyond the training data.

Practical Use Case

News summarization systems ground articles in verified sources to enhance accuracy.

RAG

Retrieval-Augmented Generation (RAG): Search and Response Generation



Definition of RAG

RAG is a method where AI searches relevant information before generating informed responses.

Practical Example

RAG searches employee handbook to answer company policy questions accurately.

Role in Language Models

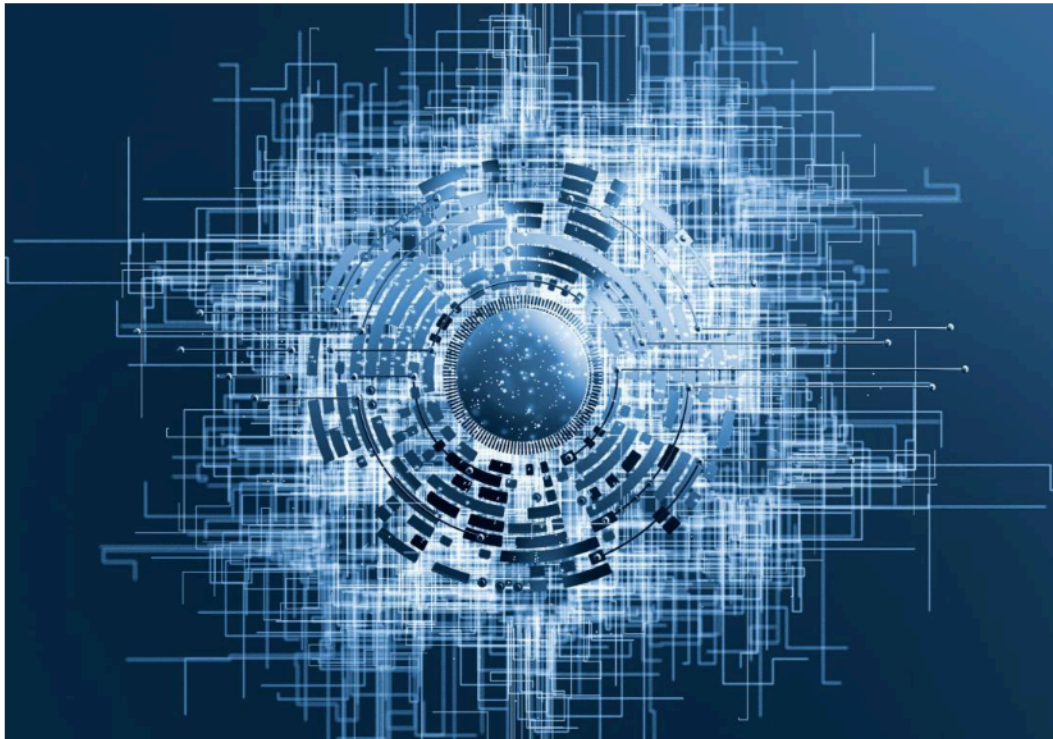
RAG integrates language model capabilities with real-time external knowledge access.

Customer Support Use Case

Customer support uses RAG to deliver precise, up-to-date product information.

Vector DB

Vector Databases: Semantic Search and Applications



Definition of Vector Databases

Vector databases store and search data based on semantic similarity instead of exact keyword matches.

Example of Semantic Search

Searching 'car problems' retrieves related documents about vehicle and automotive issues by meaning.

Role in Large Language Models

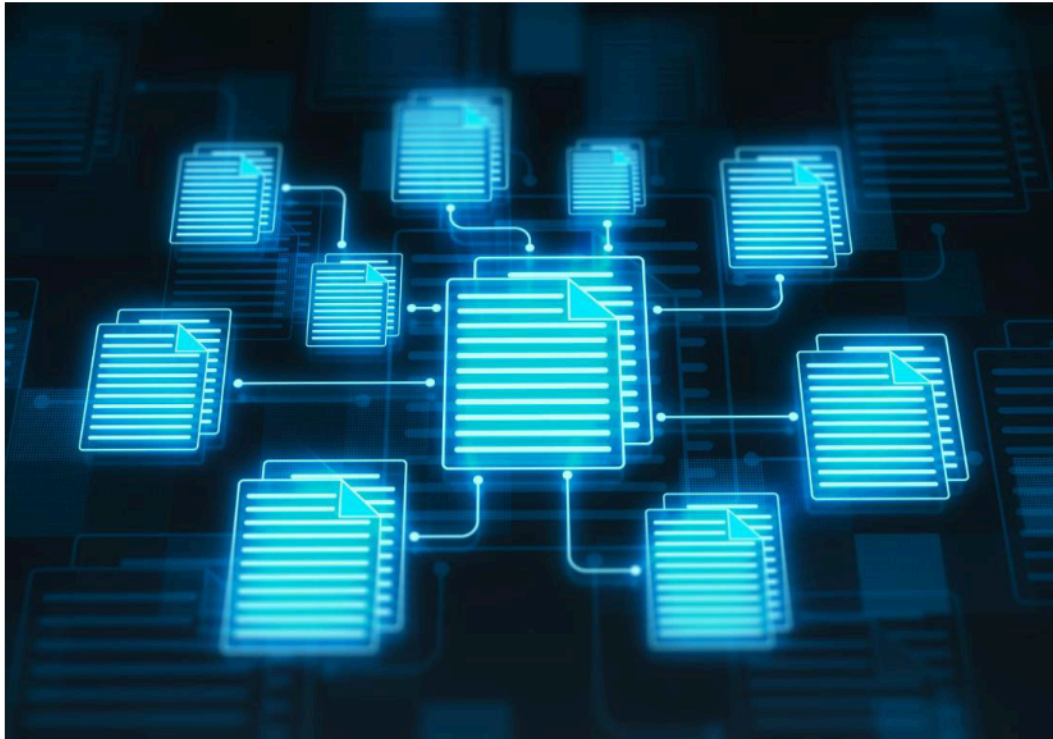
Vector databases enable LLMs to understand context and meaning, improving semantic search capabilities.

Applications in E-Commerce

E-commerce platforms use vector databases for smarter product search and personalized recommendations.

Chunking

Chunking: Document Processing for AI



Concept of Chunking

Chunking breaks large documents into smaller, manageable pieces for efficient AI processing.

Example of Document Sections

A long manual is divided into sections such as Installation, Setup, and Troubleshooting for clarity.

Role in Large Language Models

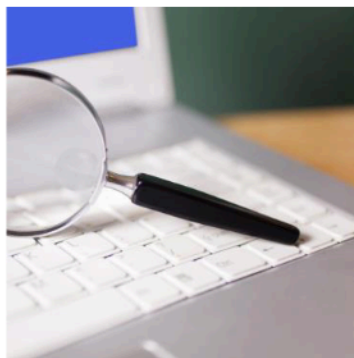
Chunking enables models to handle documents exceeding their context window size effectively.

Practical Use Case

Legal research systems use chunking to quickly locate relevant clauses in lengthy contracts.

Reranking

Reranking: Improving Search Result Relevance



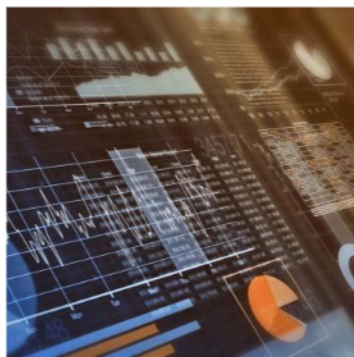
Definition of Reranking

Reranking reorders search results to prioritize the most relevant information at the top.



Role in Language Models

Reranking enhances the input quality for language models, improving their response accuracy.



Practical Use Case

Research assistants use reranking to highlight the most relevant academic papers effectively.

Evaluation & Safety Concepts

Bias

Bias: Fairness and Detection in AI Systems

Unfair Treatment in AI

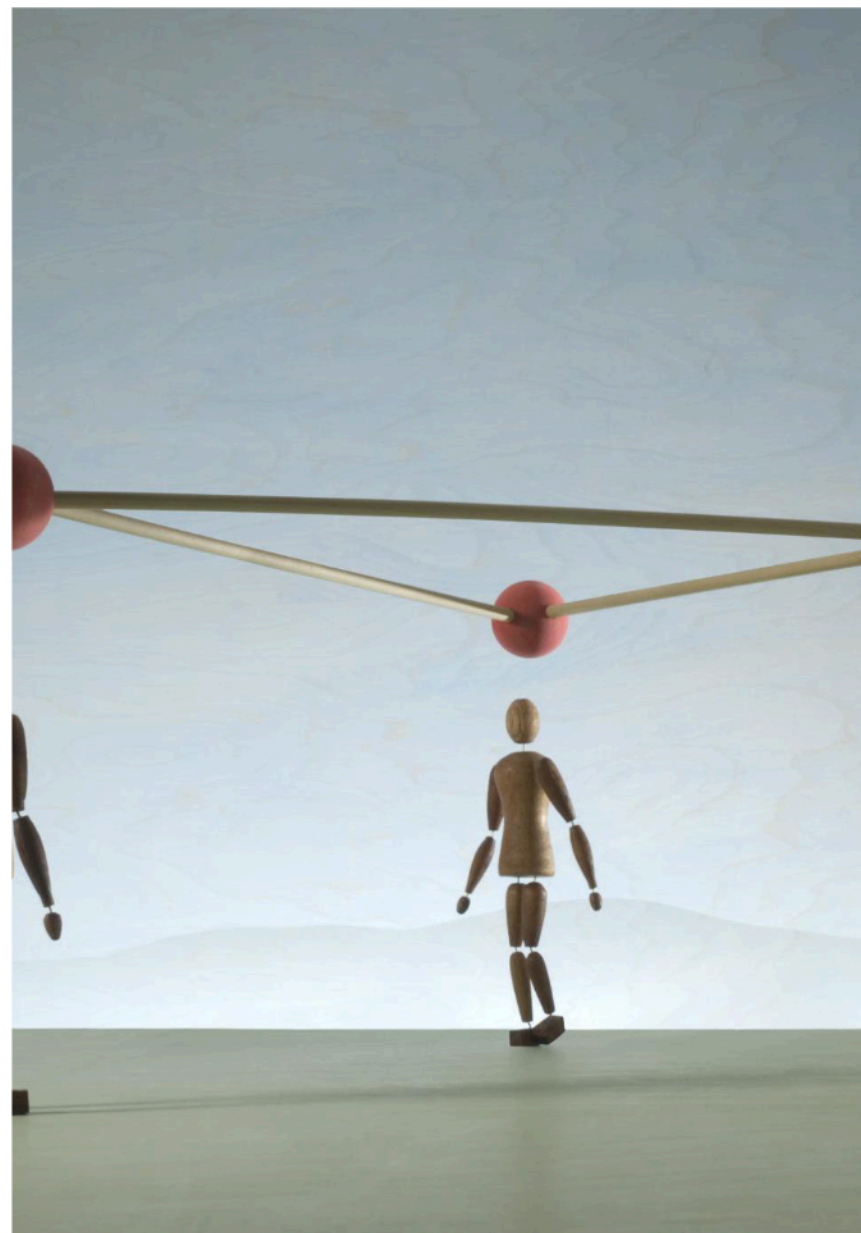
AI systems can embed prejudices, leading to unfair treatment of certain groups in decisions like hiring.

Bias in Language Models

Large language models may perpetuate societal biases present in their training data, impacting fairness of outputs.

Bias Detection in Use Cases

Content moderation systems require bias detection mechanisms to ensure fair treatment of all users.



Guardrails

Guardrails: Safety Measures and Responsible AI



Definition of Guardrails

Guardrails are safety measures preventing AI from producing harmful or inappropriate content.

Examples of Guardrails

Guardrails block AI from generating dangerous instructions or sharing personal information.

Role in Large Language Models

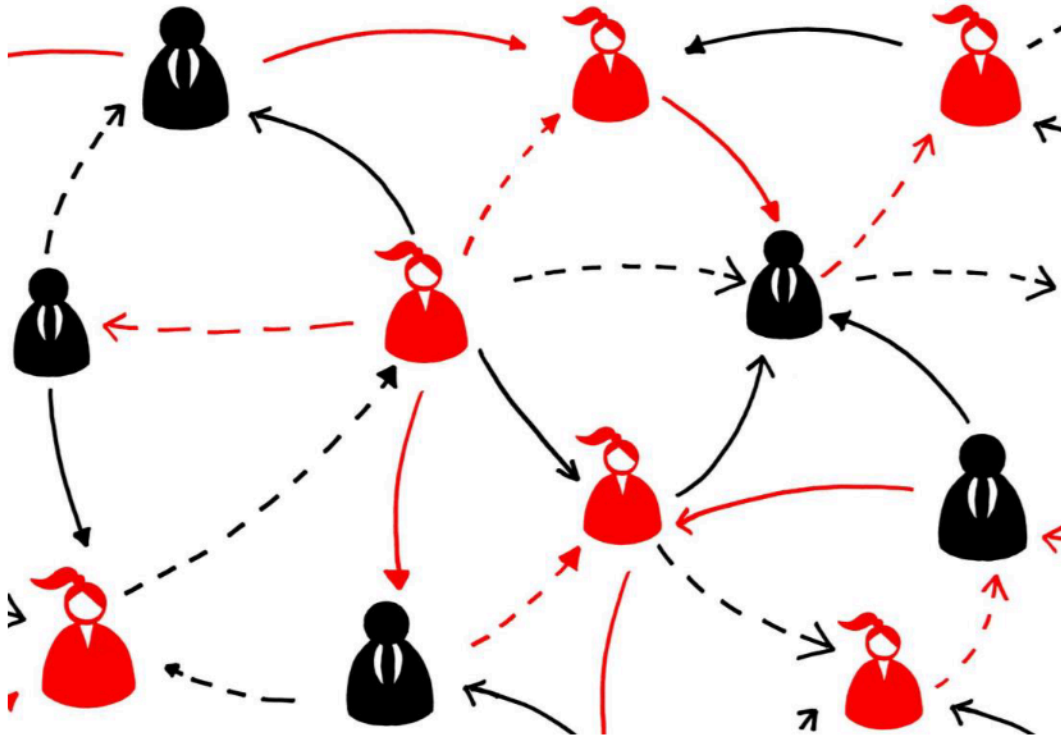
Guardrails ensure responsible AI behavior and prevent misuse during deployment.

Use Case in Education

Educational AI uses guardrails to keep content appropriate for different age groups.

Explainability

Explainability: Understanding AI Decisions



Definition of Explainability

Explainability is the ability to understand and clearly explain how an AI system reaches a decision.

Example of AI Explanation

A loan system explains approvals based on factors like credit score, income, and debt ratio percentages.

Role in Large Language Models

Explainability is crucial for trust and accountability in high-stakes AI decisions, especially in LLMs.

Healthcare Use Case

AI systems in healthcare must explain diagnoses to doctors, ensuring validation and building trust.

Conclusion

Generative AI Scope

Generative AI includes foundational principles and advanced techniques across multiple domains.

Importance of Understanding

Grasping AI essentials is crucial for developing effective and responsible AI applications.

And the glossary is endless: Link to stay updated

CLICK HERE 